

Sati-Loop

Design Specification for an Alignment-Aware Contemplative Cybernetic Platform

Version 1.0

Joy Bose

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| *A closed-loop system that fades itself into obsolescence.*

Companion to:

*Bose, J. (2026). The Contemplative Alignment Problem:
Reward Misspecification in Closed-Loop Meditation Systems.
PsyArXiv. <https://osf.io/preprints/psyarxiv/>*

Abstract

Sati-Loop is an open-source hardware and software platform for contemplative neurotechnology, designed from the ground up to avoid the alignment failures identified in existing meditation devices. Most closed-loop meditation systems optimise for measurable proxy signals such as calm EEG, HRV coherence, or breathing resonance, and reward the practitioner for producing those signals. This creates a reinforcement learning loop in which the practitioner's brain, as an adaptive optimiser, may learn dullness, suppression, or performance mimicry rather than genuine contemplative skill.

Sati-Loop addresses this through a three-layer architecture. The attentional scaffold layer (Layer 1) detects mind-wandering onset and delivers a minimal cue, with systematic scaffold withdrawal and transfer to unassisted practice as the primary success criterion. The autonomic support layer (Layer 2) provides HRV-guided breath pacing and state-responsive ambient audio to create physiological conditions for practice without directly rewarding proxy brain states. The experimental modulation layer (Layer 3) provides a research-only module for transcutaneous auricular vagus nerve stimulation (taVNS), constrained by hardware interlocks that prevent it from being connected to the classifier reward loop.

This document provides a complete design specification: sensor stack, compute architecture, signal processing pipeline, classifier design, feedback and intervention protocols, safety constraints, bill of materials for two prototyping paths, and a transfer test protocol. It is intended to enable independent replication and to establish the architecture before hardware implementation.

1. Design Philosophy

1.1 The Problem This Design Solves

Every design decision in Sati-Loop follows from a single diagnosis: existing meditation devices reward proxy physiological states rather than genuine contemplative development. The theoretical basis for this diagnosis is the accompanying preprint (Bose, 2026), which formalises the problem using AI alignment vocabulary.

The core problem is Goodhart's Law applied to inner life. When a measurable neurophysiological signal becomes the target of a feedback loop, the practitioner's brain will find the most efficient path to producing that signal. Calm EEG can be produced by genuine settled awareness, by mild dissociation, by pleasant dullness, or by a specific postural or muscular configuration that creates the target waveform as an artefact. A device that rewards all of these equally is not training meditation. It is training proxy production.

Sati-Loop is designed to avoid this by doing three things that no existing consumer device does. First, it uses negative-only feedback: the only signal the device sends is a cue when distraction is detected. There is no reward for calm. Second, it measures success not by in-session

performance but by whether the skill transfers to practice without the device. Third, it systematically withdraws its own support as skill develops, making itself unnecessary over time.

1.2 Non-Negotiable Design Rules

The following constraints are hard-coded into the architecture and cannot be overridden by configuration:

- Detect probable distraction onset only. Do not attempt to detect equanimity, insight, compassion, or any Tier 3 or Tier 4 contemplative state.
- Deliver one minimal cue when self-correction window expires. Never deliver continuous feedback or reward signals.
- No positive reinforcement for calm EEG or any other proxy signal. The device never tells the practitioner they are doing well.
- No scores, dashboards, session ratings, or gamification of any kind.
- Scaffold withdrawal is mandatory. The intervention must decrease as skill develops.
- Transfer to unassisted practice is the only success criterion. In-session performance is not a success criterion.
- No electrical or magnetic brain stimulation in Layer 1 or Layer 2. Stimulation is confined to Layer 3 (research only) with hardware interlocks.
- All data remains on the user's device unless explicit written consent is given for research upload.
- Full open-source: hardware schematics, firmware, signal processing code, protocols, and documentation.

1.3 What This System Is Not

Sati-Loop is not a meditation app. It does not teach meditation. It does not guide practice. It does not evaluate the quality of a session. It does not claim to induce any specific mental state. It does not detect enlightenment, equanimity, compassion, rigpa, or any other Tier 3 or Tier 4 contemplative state as defined in the accompanying paper. Any future version or fork that makes such claims violates the design philosophy and should not use the Sati-Loop name.

2. System Architecture

2.1 Three-Layer Overview

The system is divided into three operationally distinct layers. Each layer has explicit constraints on what it can and cannot do. The layers communicate upward (sensing to estimation to intervention) but the intervention layer is never permitted to feed back into the classifier reward signal.

Layer	Name	Function	Default State
1	Attentional Scaffold	Mind-wandering detection, minimal bone-conduction cue, scaffold withdrawal, transfer testing	Active
2	Autonomic Support	HRV-guided breath pacing, state-responsive ambient audio	Optional, user-enabled

Layer	Name	Function	Default State
3	Experimental Modulation	taVNS for parasympathetic support; tDCS/tACS research module	Off by default, research use only

Layer 3 requires explicit activation at session start. It cannot be enabled from the classifier or from Layer 1 or Layer 2 logic. Hardware interlocks prevent the stimulation current from being controlled by the ML pipeline.

2.2 The Critical Architectural Constraint

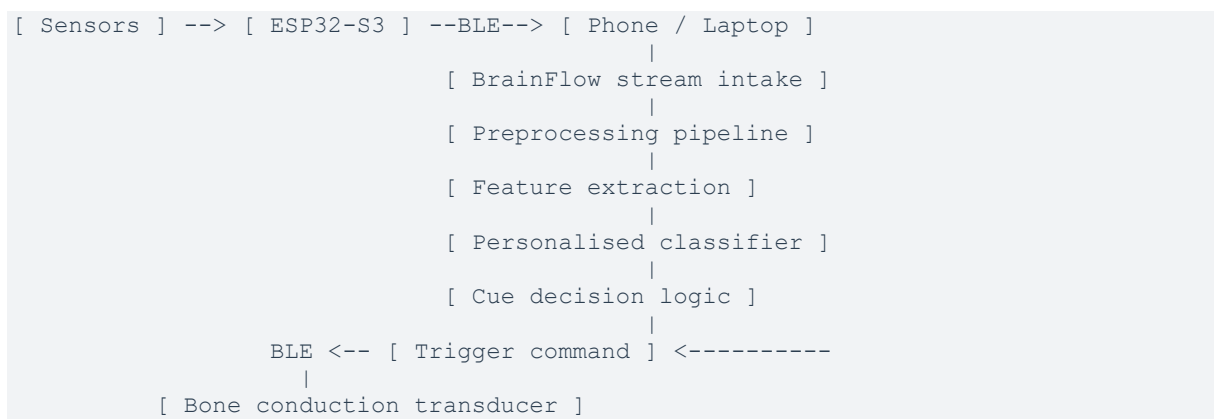
The most important design rule in the entire system, stated precisely:

The intervention engine must never optimise directly against the classifier output. Stimulation parameters, audio parameters, and cue parameters are set by the practitioner or researcher at session initialisation. They are not adjusted by the ML model during a session. The classifier detects states and triggers interventions, but it does not tune intervention intensity toward maximising any biomarker.

This constraint is what distinguishes Sati-Loop from every existing closed-loop neurofeedback system. Every other system creates a loop where the intervention is tuned to maximise the target signal. Sati-Loop explicitly breaks this loop. The classifier informs the intervention trigger. It does not control intervention magnitude.

2.3 Compute Architecture

The system uses a hybrid compute model. Sensor data is acquired and buffered on an embedded microcontroller (ESP32-S3), streamed via BLE to a smartphone or laptop running the signal processing and classification pipeline, and cue commands are sent back to the ESP32 for delivery via the bone conduction transducer.



On-device processing on the ESP32 handles sensor polling, timestamping, buffering, and BLE streaming. The ESP32 also runs a simple local failsafe detector for gross agitation and dullness

that can trigger a cue if the phone connection is lost. All ML classification runs on the phone or laptop for flexibility and power.

3. Sensing Layer

3.1 Design Philosophy for Sensing

The sensing layer uses four modalities because meditation states are not purely neural. They are attentional, autonomic, respiratory, and postural. Single-signal systems are fundamentally impoverished as state estimators for contemplative practice. Each modality contributes something that the others cannot: EEG provides temporal resolution on neural dynamics, HRV provides autonomic state, respiration provides breath patterns directly relevant to practice, and IMU provides postural information critical for distinguishing dullness from stillness.

Critically, multimodal integration provides partial de-confounding of the states that are most problematic for proxy-based systems. EEG alone cannot distinguish subtle laxity from genuine stillness. But when EEG shows elevated theta and IMU shows a forward head droop of more than ten degrees and respiration has slowed and deepened, the combination points toward laxity rather than genuine samadhi. This is not a solved problem but multimodal sensing makes it more tractable than any single signal.

3.2 Two Prototyping Paths

The system is designed to run on two different hardware stacks. The entry-level path uses a Muse S headband for EEG, supplemented by an external ESP32-S3 module for the other sensors. The research path uses an OpenBCI Cyton board for full 8-channel EEG. The software pipeline switches between them by changing a single BoardId parameter in BrainFlow. This allows researchers and makers to start immediately at low cost and upgrade to the research stack without rewriting any software.

Parameter	Entry Path (Muse S)	Research Path (OpenBCI Cyton)
EEG channels	4 (Fp1, Fp2, TP9, TP10)	8 (Fp1, Fp2, Fz, Cz, TP9, TP10, C3, C4)
EEG resolution	12-bit, ~256 Hz	24-bit, 250 Hz (ADS1299 AFE)
EEG noise floor	~1.0 uV RMS	~0.5 uV RMS
BrainFlow BoardId	MUSE_S_BOARD	CYTON_BOARD
EEG cost	~USD 380	~USD 999
Other sensors	External ESP32 module	External ESP32 module
Total approx cost	~USD 450	~USD 1,300
Use case	Prototyping, personal practice	Research, publication-grade data

3.3 Sensor Stack Specification

EEG

Electrode placement targets the frontal and temporoparietal regions. Fp1 and Fp2 provide frontal coverage relevant to default mode network activity and attentional state. TP9 and TP10 serve as reference and provide temporal coverage. On the research path, Fz and Cz provide midline theta coverage, and C3 and C4 provide central coverage for the full 8-channel configuration.

Dry electrodes are required for wearable use. The OpenBCI dry comb electrodes are preferred for the research path. Florida Research Instruments Ag/AgCl dry electrodes are an alternative with better signal quality at higher cost. Gel electrodes produce better signal but are impractical for repeated daily use.

PPG / HRV

Property	Specification
Component	MAX30102 breakout (Adafruit 4046 or equivalent)
Interface	I2C to ESP32-S3
Placement	Ear clip or finger clip
Sample rate	50 Hz (raw IR + red channels)
Derived metrics	Inter-beat intervals, RMSSD, pNN50, LF/HF ratio
HRV window	30-second rolling window
Cost	~USD 9

Respiration

Property	Specification
Component	Conductive stretch sensor (Adafruit 1171 or equivalent)
Interface	Voltage divider to ESP32-S3 ADC (GPIO 34)
Placement	Lower chest or upper abdomen
Sample rate	50 Hz after ADC
Derived metrics	Breath rate, tidal depth, irregularity coefficient, breath-by-breath variability
Cost	~USD 10

IMU / Posture

Property	Specification
Component	BNO085 9-axis IMU (Adafruit 4754) or MPU6050 (lower cost)
Interface	I2C to ESP32-S3
Placement	Rear of headband, base of skull
Sample rate	100 Hz
Derived metrics	Head pitch, roll, yaw; tilt velocity; micro-movement jerk; postural drift

Property	Specification
BNO085 advantage	On-board sensor fusion reduces ESP32 compute load
Cost	~USD 12-15

Optional: Galvanic Skin Response

GSR / EDA is not included in version 1 but is worth adding in version 2. The Adafruit GSR sensor costs approximately USD 10 and provides a strong signal for gross agitation detection. It is excluded from version 1 to keep the hardware scope manageable and because HRV and EEG together cover the agitation detection use case adequately.

3.4 ESP32-S3 Wiring Summary

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ESP32-S3 I2C bus (GPIO 21 SDA, GPIO 22 SCL):
  MAX30102 PPG      ---> I2C address 0x57
  BNO085 IMU        ---> I2C address 0x4A

ESP32-S3 ADC:
  Stretch sensor    ---> GPIO 34 (ADC1_CH6)
  Voltage divider: sensor in series with 10k resistor, 3.3V supply

ESP32-S3 UART (for OpenBCI Cyton path):
  Cyton TX          ---> ESP32 RX (GPIO 16)
  Cyton RX          ---> ESP32 TX (GPIO 17)

ESP32-S3 I2S (bone conduction output):
  MAX98357A BCLK    ---> GPIO 27
  MAX98357A LRC     ---> GPIO 25
  MAX98357A DIN     ---> GPIO 26

Power:
  3.7V LiPo --> TP4056 charger --> ESP32 VIN
  Boost converter 5V for OpenBCI Cyton supply

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Pin conflict note (v1.1 correction): GPIO 22 is used for I2C SCL. The I2S DIN line must not share this pin. Assign I2S as shown above: BCLK to GPIO 27, LRC to GPIO 25, DIN to GPIO 26. On the ESP32-S3-DevKitC-1 these pins are free. Verify against your specific board variant before assembly.

4. Signal Processing Pipeline

4.1 Data Rates and Windowing

Modality	Sample Rate	Window	Step
EEG	250-256 Hz	2.0 seconds	0.5 seconds
PPG / HRV	50 Hz (raw); 30s rolling window for HRV	30 seconds	1 beat
Respiration	50 Hz	10 seconds	1 second
IMU	100 Hz	2.0 seconds	0.5 seconds

4.2 Preprocessing

EEG

- Notch filter: 50 Hz and 60 Hz (2nd order IIR, both to handle international use)
- Bandpass: 0.5 to 40 Hz (Butterworth, order 4)
- Common average reference (CAR)
- Amplitude artifact rejection: discard windows with any channel exceeding +/- 100 uV
- Slope artifact rejection: discard windows with inter-sample delta exceeding 50 uV
- Eye blink detection: frontopolar channels showing characteristic blink morphology flagged as missing data, not as distraction
- Jaw clench detection: broadband power spike across all channels flagged as artifact

PPG

- Low-pass filter: 5 Hz
- Adaptive peak detection for inter-beat intervals
- Ectopic beat removal: IBIs outside 0.3x to 2x rolling median removed
- HRV metrics computed over 30-second rolling window: RMSSD, pNN50, LF/HF ratio via FFT on IBI series

Respiration

- Low-pass filter: 2 Hz (removes movement noise)
- Zero-crossing detection for breath cycle identification
- Derived metrics: rate in breaths per minute, tidal amplitude, coefficient of variation for irregularity

IMU

- Complementary filter combining accelerometer and gyroscope (or BNO085 fusion output)
- Derived metrics: absolute tilt angle, tilt velocity, jerk (RMS of acceleration derivative), low-frequency postural drift

4.3 Feature Extraction

Features are extracted on each 2-second window stepped at 0.5 seconds. The feature vector is multimodal, combining all four sensing streams.

Modality	Features
EEG spectral	Absolute and relative band power: delta (0.5-4 Hz), theta (4-8 Hz), alpha (8-12 Hz), beta (12-30 Hz), gamma (30-40 Hz)
EEG spatial	Frontal alpha asymmetry (Fp1 minus Fp2 alpha power), frontal midline theta (Fz), theta/beta ratio, frontal/parietal alpha ratio
EEG complexity	Shannon entropy per channel, Hjorth mobility, Hjorth complexity
HRV	RMSSD, pNN50, LF power, HF power, LF/HF ratio, mean IBI
Respiration	Rate, tidal depth, coefficient of variation, breath-HRV coupling (respiratory sinus arrhythmia phase coherence)

Modality	Features
IMU	Pitch angle (head droop), pitch velocity, jerk RMS, postural stability index (low-frequency spectral power of tilt)
Multimodal	Cross-correlation of frontal alpha and respiration (certain meditative states show coupling)

4.4 State Classification

Feature speed constraints: Layer 1 mind-wandering detection must rely exclusively on high-speed EEG features. PPG-derived HRV metrics have a physiological and computational latency of several seconds and cannot capture the micro-moment of distraction onset. If HRV is included in the Layer 1 classifier, cues will consistently fire after the practitioner has already drifted into a deep narrative loop. HRV, respiration, and IMU metrics are relegated to Layer 2 and Layer 3 for slow-moving baseline state estimation (agitation and dullness). The Layer 1 classifier input vector should contain only EEG spectral and complexity features.

Calibration Phase (Weeks 1 and 2)

The classifier is personalised to each user during a two-week passive sensing baseline. During this phase no cues are delivered. At random intervals (mean 3 minutes, jitter +/- 90 seconds), the app delivers a brief auditory tone and presents a three-option phenomenological probe on screen: Settled, Wandering, or Unclear. The user taps their response within five seconds.

Each probe generates a labelled sample: the feature vector from the two-second window ending at the probe time, labelled with the user's response. Approximately 80 to 120 labelled samples are collected over the two-week baseline. This constitutes the personalised training set.

Classifier

- Algorithm: Random Forest (100 trees) or XGBoost, binary classification: Wandering vs Not-Wandering
- Class balancing: SMOTE oversampling of the minority class (typically Wandering)
- Validation: leave-one-session-out cross-validation within the calibration period
- Output: probability p in $[0, 1]$ of mind-wandering
- Initial threshold T : 0.70 (tuned per user based on calibration performance)
- Unclear responses: excluded from training; used to estimate label uncertainty

Real-Time Inference

During active sessions the classifier runs on each 0.5-second step, producing a new $p(\text{wandering})$ estimate every 500ms. The cue decision logic is:

```
if p > T:
    and time_since_last_cue > min_interval (default 3 seconds):
        and self_correction_not_detected:
            trigger_cue()
            log_event(timestamp, p, feature_vector)
```

4.5 Adaptive Scaffold Withdrawal

The scaffold withdrawal mechanism is the most important adaptive component of the system. It is designed to make the device progressively less necessary as the practitioner's metacognitive skill develops.

After each session the system computes the mean cue rate (cues per minute). If the 7-day rolling mean cue rate decreases for 5 consecutive sessions, the threshold T is increased by 0.02 (making the classifier less sensitive) and the minimum cue interval is increased by 0.5 seconds. Both adjustments are capped: T at 0.92, interval at 8 seconds.

Weekly transfer test sessions (described in Section 7) provide an additional check. If the transfer test shows that the unassisted correction gap is not improving, the withdrawal is paused and the previous parameters are restored. This prevents premature scaffold removal.

5. Feedback and Intervention

5.1 Layer 1: The Minimal Cue

The only intervention in Layer 1 is a single soft tone delivered through the bone conduction transducer when distraction is detected and the self-correction window has expired. The design of this cue is as minimal as possible: it should be noticeable without being startling, and it should carry no reward or punishment valence.

Parameter	Specification	Rationale
Transducer	Dayton Audio BCT-2 bone conduction (or equivalent)	Leaves ear canals open; practitioner remains in acoustic environment
Amplifier	MAX98357A I2S mono amplifier (Adafruit 3006)	Low noise, sufficient power for bone conduction
Tone frequency	Subtractive high-frequency roll-off in ambient soundscape, 500ms, -12dB at 4kHz. Optional fallback: 440Hz sine, 200ms, only if no ambient audio is active.	Subtractive modulation avoids habituation and autonomic startle. A fixed additive tone risks orienting reflex conditioning over repeated sessions. Removing high-frequency content from the soundscape is less jarring and carries no positive or negative valence.
Duration	200ms with 20ms fade-in and 20ms fade-out	Short enough not to disrupt, long enough to register
Maximum SPL	65 dB at transducer (hardware-limited)	Well below safe exposure thresholds for any session length
Cue shape	Single tone only; no repeating, no escalation	No urgency or reward signal embedded in the cue pattern

The device must never play a pleasant, rewarding, or congratulatory sound. No chimes for good sessions, no increasing frequencies for good states. The only audio event is the distraction cue, and it is neutral.

5.2 Layer 2: Autonomic Support (Optional)

Layer 2 is enabled by the practitioner at session start. It does not activate automatically based on classifier output, because activation itself constitutes a form of positive reinforcement. When enabled, it runs continuously at low intensity and responds to gross agitation and dullness in ways that support the physiological substrate of practice without rewarding proxy states.

HRV-Guided Breath Pacing

When Layer 2 is active and gross agitation is detected (elevated HRV LF/HF ratio, irregular respiration, elevated beta EEG), the ambient audio subtly shifts its rhythm toward the practitioner's personal cardiovascular resonant frequency, typically around 0.1 Hz or 6 breaths per minute. This is not a visual pacer or a gamified score. It is a very gentle rhythmic quality in the ambient sound that the practitioner may or may not consciously notice. The effect is to support baroreflex sensitivity and parasympathetic tone through subtle entrainment.

The resonant frequency is determined during a dedicated 10-minute calibration session run at the end of Phase A Week 1. The protocol is as follows. The app delivers a gentle auditory breath pacer at five rates in sequence: 5.0, 5.5, 6.0, 6.5, and 7.0 breaths per minute, spending 90 seconds at each rate. The system measures HRV coherence (ratio of peak power at the pacing frequency to total LF power in the 0.04 to 0.15 Hz band) during each 90-second window. The rate producing the highest coherence is stored as the personal resonant frequency and used for all subsequent Layer 2 breath pacing. If coherence does not peak clearly at any single rate, the default of 6.0 breaths per minute (0.1 Hz) is used. The calibration should be repeated at 4-week intervals as autonomic tone changes with practice.

State-Responsive Ambient Audio

The ambient audio layer (pink noise or natural soundscape, user's choice) responds to estimated state in two ways only. When gross agitation is detected, the audio very slowly and subtly shifts to a lower frequency centre, which is acoustically calming. When gross dullness is detected, the audio shifts to a slightly brighter texture. These shifts are smooth, slow (over 30 seconds), and subtle. They do not constitute a reward or punishment signal. They create ambient conditions more conducive to the relevant corrective direction.

5.3 Layer 3: Experimental Modulation (Research Only)

Layer 3 is the taVNS module. It is off by default, requires explicit activation at session initialisation, and is subject to hardware interlocks that prevent the stimulation current from being controlled by the classifier or by any Layer 1 or Layer 2 logic.

Why taVNS Rather Than tDCS or tACS

tDCS and tACS deliver electrical current directly to the scalp targeting cortical regions. Their effect is to modulate the firing threshold of cortical neurons. If the classifier were permitted to adjust tDCS or tACS parameters to maximise a brain state proxy, the result would be direct neural state forcing, which is the highest-risk form of the contemplative alignment problem. The system would be both measuring and aggressively pushing brain states toward a proxy target. This is the failure mode the paper calls the alignment problem squared.

taVNS acts through a different pathway. Stimulation is delivered to the auricular branch of the vagus nerve via a tragus clip on the left ear. The vagus nerve projects to the nucleus tractus solitarius in the brainstem, which regulates the autonomic nervous system. The effect is bottom-up parasympathetic upregulation, reducing sympathetic arousal and supporting the physiological conditions for settled practice. It does not directly modulate cortical firing patterns or brain state content. This is categorically different from tDCS/tACS, and it is why taVNS can be included in a research module without triggering the full alignment problem.

taVNS Parameters

Parameter	Value	Source
Current amplitude	0.5 to 2.0 mA (set at session start, not adjusted during session)	Standard research range for auricular taVNS
Pulse frequency	25 Hz	Most common in published research; supports parasympathetic activation
Pulse width	250 microseconds	Standard for auricular stimulation
Duty cycle	30 seconds on, 30 seconds off (fixed)	Prevents accommodation; mirrors clinical protocols
Electrode	Left tragus clip, inner surface	Auricular vagal innervation is primarily at the tragus and concha
Maximum session	30 minutes of active stimulation	Conservative limit; most research protocols use 20-30 minutes
Contraindications	Cardiac pacemaker, implanted neural devices, pregnancy, active epilepsy	Standard taVNS exclusions

Hardware Interlocks

Three hardware-level constraints are mandatory for the taVNS module:

- **Electrical isolation:** The taVNS current generation circuit must run on a battery rail that is physically separate from the sensing electronics. Optocouplers must isolate all digital control lines. This prevents switching noise from contaminating the EEG and prevents any possibility of leakage current through the sensing array.
- **Manual current setting:** The stimulation amplitude is set by the researcher or practitioner at session initialisation via a physical potentiometer or a hardware-level DAC input that cannot be written to by the classification software during a running session. Software can read the current setting but cannot write to it.
- **Hardware kill switch:** A physical push-button must be accessible during the session that immediately terminates stimulation by opening the circuit. Software cannot override this switch.

- **Time-division multiplexing (TDM):** When taVNS is active, the 2 mA rectangular pulse at 25 Hz introduces electromagnetic interference that can saturate the microvolt-level EEG input amplifiers. The firmware must implement TDM gating: stimulate for 400ms, pause for 100ms, repeat. EEG feature extraction runs exclusively during the 100ms silent windows. The signal pipeline must be configured to reject windows that overlap with stimulation-active periods. This constraint must be enforced at firmware level and documented in the firmware TDM gating module.

tDCS and tACS are not included in any version of Sati-Loop designed for personal or consumer use. They may be added to a separate research module in a future version, with IRB approval, full safety protocols, and explicit exclusion from the classifier reward loop. This document does not specify tDCS/tACS hardware.

6. Safety and Ethics Constraints

6.1 Hard-Coded Safety Rules

The following rules are enforced at the firmware level and cannot be overridden by application software or configuration files:

- No electrical stimulation output from Layer 1 or Layer 2 hardware under any circumstances.
- Maximum bone conduction SPL: 65 dB. Hardware amplifier gain is fixed at a level that enforces this limit regardless of software volume settings.
- Session duration limit: 90 minutes. The device automatically ends the session and requires a 15-minute break before restarting.
- No continuous positive reinforcement audio. The firmware rejects any audio command lasting more than 500ms that is triggered by a positive state classification.
- taVNS stimulation terminates immediately if the hardware kill switch is pressed. This cannot be overridden.

6.2 Ethical Constraints

The following constraints are implemented in software and documented in all user-facing materials:

- No claims about meditation quality, depth, or progress. The app displays only: session duration, number of distraction cues delivered (declining is good), and transfer test results.
- No claims about physical or mental health outcomes. The device is a research tool and a training aid, not a medical device.
- No Tier 3 or Tier 4 claims. The device does not detect, measure, or induce equanimity, insight, compassion, rigpa, stream-entry, or any attainment described in the contemplative taxonomy in the accompanying paper.
- Adverse effect monitoring: Users are asked weekly to complete a brief questionnaire covering depersonalisation, emotional numbing, mood instability, and sleep disturbance. This is presented separately from session performance data. If a user reports two or more adverse effects in three consecutive weeks, the app recommends consulting a teacher or clinician and automatically reduces intervention intensity.

- Data privacy: All raw sensor data and phenomenological reports are stored locally on the user's device in an encrypted local database. Upload to any external server requires explicit opt-in with full informed consent describing what data is collected and how it will be used.

6.3 Disclaimer

Sati-Loop is an experimental research platform. It is not a medical device and has not been evaluated by any regulatory body. It is not a substitute for instruction from a qualified meditation teacher. Users with epilepsy, cardiac conditions, or implanted electrical devices must not use Layer 3 (taVNS). Users experiencing significant psychological distress should consult a clinician before using any neurofeedback system. The authors make no claims about the efficacy, safety, or reliability of this system for any specific individual or use case.

7. Transfer Test Protocol

7.1 The Primary Hypothesis

All other metrics are secondary. The only question that matters is:

Does the interval Delta t between distraction onset and conscious self-correction decrease in unassisted practice (device absent) after training with the device? If Delta t decreases only with the device active, the system has produced dependency, not skill.

This is a falsifiable hypothesis. The transfer test is the empirical procedure for testing it.

7.2 Study Protocol

Phase	Duration	Device State	Procedure
Phase A: Baseline	Weeks 1-2	Passive sensing only; no cues	Calibration phase. Measure Delta t via phenomenological probes. Establish personal classifier baseline.
Phase B: Training	Weeks 3-8	Active closed-loop with scaffold withdrawal	Full Layer 1 active. Weekly Monday sessions without device to measure unassisted Delta t.
Phase C: Transfer test	Weeks 9-10	Device absent	Practice without device. Measure Delta t via app-timed phenomenological probes only (no EEG feedback).
Phase D: Follow-up	Month 3 and Month 6	Device absent	Optional long-term transfer check. Determines whether gains persist over time.

7.3 Control Condition

A sham control condition is required for any published study. The sham device delivers cues at random intervals uncorrelated with detected distraction states, using the same hardware and

cue tone. Sham participants cannot distinguish their device from the active device. This controls for expectation effects, interruption effects, and general attentional priming.

7.4 Success Criterion

The study is successful if and only if the Phase C unassisted Delta t is significantly lower than the Phase A baseline Delta t (paired t-test or Wilcoxon signed-rank, $\alpha = 0.05$), and if the effect size is maintained at Phase D follow-up. In-session Delta t improvement during Phase B is not a success criterion. It is expected and uninformative about transfer.

8. Bill of Materials

8.1 Entry-Level Prototype (~USD 450)

Component	Part / Source	Cost (USD)	Notes
EEG headband	Muse S Gen 2 (InteraXon)	380	BrainFlow MUSE_S_BOARD; dry electrodes; BLE
Microcontroller	ESP32-S3-DevKitC-1 (Espressif / Mouser)	20	BLE, WiFi, sufficient SRAM
PPG sensor	MAX30102 breakout (Adafruit 4046)	9	I2C; ear clip included
Respiration sensor	Conductive stretch sensor (Adafruit 1171)	10	Voltage divider to ADC
IMU	BNO085 breakout (Adafruit 4754)	15	I2C; on-board fusion
Bone conduction transducer	Dayton Audio BCT-2 (Parts Express 295-238)	13	0.5W; attach to headband
Audio amplifier	MAX98357A I2S (Adafruit 3006)	10	3.7W mono; I2S input
LiPo battery	1000 mAh 3.7V (Adafruit 1578)	10	8+ hours operation
Charger	TP4056 USB-C module (generic)	3	Charge while not in use
Misc	Wires, connectors, fabric headband, 3D print	20	Assembly materials
Total		~490	Sufficient for software validation and personal use

8.2 Research-Grade Prototype (~USD 1,300)

Component	Part / Source	Cost (USD)	Notes
EEG board	OpenBCI Cyton (openbci.com)	999	8 channels; 24-bit ADS1299; open ecosystem
EEG electrodes	OpenBCI dry comb electrodes x8	200	Fp1, Fp2, Fz, Cz, TP9, TP10, C3, C4
Microcontroller	ESP32-S3-DevKitC-1	20	Co-processor for other sensors + BLE
PPG sensor	MAX30102 breakout	9	As above
Respiration sensor	Conductive stretch sensor	10	As above
IMU	BNO085 breakout	15	As above
Bone conduction transducer	Dayton Audio BCT-2	13	As above
Audio amplifier	MAX98357A I2S	10	As above
LiPo battery	1000 mAh 3.7V	10	As above
Charger	TP4056 USB-C module	3	As above
5V boost converter	Adafruit 2030 or equivalent	8	Powers Cyton from LiPo
Misc	Wires, connectors, enclosure, 3D print	30	Assembly materials
Total		~1,327	Publication-grade EEG; recommended for research studies

8.3 Optional: taVNS Module (Layer 3, Research Only, ~USD 80 DIY)

Component	Part / Source	Cost (USD)	Notes
Constant current source IC	XTR111 or Howland pump circuit (discrete)	15	Isolated constant current output
Optocouplers	PC817 x2 (generic)	3	Digital isolation from sensing electronics
Isolated battery rail	Separate 9V or LiPo (100 mAh)	10	Physically separate from main battery
Tragus electrode clip	Custom or sourced from TENS supplier	15	Stainless steel; left ear tragus
Manual potentiometer	10k linear pot with knob	5	Sets stimulation current; not software-controlled
Kill switch	Normally-closed push-button	3	Hardware interrupt; cannot be overridden

Component	Part / Source	Cost (USD)	Notes
PCB fabrication	OSHPark or JLCPCB	25	Custom PCB with isolation barrier
Misc	Resistors, capacitors, connectors	5	
Total		~81	DIY; requires electronics experience and safety testing before use

9. Known Unknowns and Limitations

This specification is more coherent than the evidence base currently supports in certain areas. The following limitations and open questions should be disclosed to any user, builder, or researcher working with this platform. This section is not defensive hedging. Honest acknowledgment of unknowns increases rather than decreases scientific credibility, and prevents overclaiming that would undermine the project's intellectual standing.

- **Classifier reliability is unvalidated.** The Random Forest classifier architecture, threshold values, and feature weights specified here are reasonable initial heuristics, not empirically validated parameters. The specific values ($T = 0.70$, $\tau = 3.0$ seconds, threshold increment of 0.02 per week) are proposed baselines. They should be treated as starting points for empirical calibration, not as established optima.
- **Phenomenological label quality is uncertain.** The neurophenomenological calibration depends on accurate self-report from the practitioner. Beginners frequently misidentify subtle laxity as settled awareness, which is precisely the confusion the paper identifies as a Tier 3 measurement problem. Training labels from beginners will therefore have systematic biases that the classifier will inherit. The system has no way to detect or correct for this independently of phenomenological training for the user.
- **Transfer is hypothesised, not demonstrated.** The central claim of the design, that attentional scaffold withdrawal produces transferable metacognitive skill, is a well-motivated hypothesis with theoretical and indirect empirical support. It has not been demonstrated in a controlled trial with this specific architecture. The transfer test protocol in Section 7 is designed to test this hypothesis, not to confirm it.
- **Individual variability is high.** EEG signatures of mind-wandering are highly individual, are affected by sleep quality, medication, emotional state, and practice history, and change over the course of a session. The personalised calibration approach addresses some of this variability, but individual differences may make the classifier unreliable for some users regardless of calibration quality.
- **Cue disruption risk is unquantified.** A cue delivered at the wrong moment, during a brief period of genuine settling for example, could disrupt a fragile state and itself cause distraction. The 3-second confirmation window is designed to reduce false positives, but false positives will still occur. The system has no way to know whether any given cue was helpful or harmful. User report is the only available signal.
- **Long-term effects of scaffold withdrawal are unknown.** The scaffold withdrawal schedule is designed to prevent dependency. Whether it does so in practice, and whether the rate of withdrawal is appropriate for practitioners at different stages of

development, is not known. The adverse effects monitoring protocol in Section 6.2 is the primary safety net for detecting problems, but it relies on user self-report and cannot detect all forms of subtle harm.

13. Repository Structure

sati-loop/ README.md CONTRIBUTING.md LICENSE	Project overview and quick-start How to contribute CERN-OHL-S (hardware), MIT (software)
paper/ preprint_link.md	Link to PsyArXiv preprint (Bose, 2026)
hardware/ Hardware_Architecture_v1.md BOM_entry.md BOM_research.md BOM_tavns.md schematics/ pcb/ STL/	This document Entry-level BOM (~USD 450) Research-grade BOM (~USD 1,300) Optional taVNS module (~USD 80) KiCad schematic files (in progress) Gerber files for custom PCBs (in progress) 3D-printable headband and enclosure files
firmware/ esp32/ main.cpp sensors.cpp audio.cpp ble_service.cpp signal_pipeline/ preprocessing.py features.py classifier.py scaffold.py cue_logic.py	ESP32-S3 Arduino / ESP-IDF firmware Main loop: sensor polling, BLE streaming MAX30102, BNO085, stretch sensor drivers Bone conduction cue generation BLE Nordic UART Service implementation BrainFlow intake, filtering, artifact rejection Feature extraction for all modalities Personalised RF/XGBoost classifier Adaptive withdrawal logic Cue decision and trigger
app/ calibration/ session/ transfer_test/ adverse_effects/	Mobile app (Flutter, in progress) Phase A probe delivery and label collection Active session interface Weekly unassisted session protocol Weekly safety questionnaire
protocol/ Transfer_Test_Protocol.md Calibration_Protocol.md taVNS_Safety_Protocol.md	Full experimental protocol Baseline and labelling procedure Constraints and contraindications for Layer 3
docs/ Assembly_Guide.md Signal_Pipeline_Guide.md Philosophy.md	Step-by-step build instructions How to run the Python pipeline Design philosophy and alignment framework

11. Non-Optimization Constraints

This section constitutes the anti-Goodhart engineering layer of the specification. It defines what the system must never do, stated as hard constraints rather than design preferences. Any fork, derivative, or implementation claiming compatibility with this specification must satisfy all constraints in this section.

The system MUST NOT do any of the following, regardless of user request, configuration setting, or downstream fork.

10.1 Prohibited Optimisation Targets

- Optimise directly for alpha amplitude, theta power, HRV coherence, or any other single biomarker as a session score.
- Display calmness scores, mindfulness ratings, meditation depth indices, or any composite metric derived from proxy signals.
- Maximise session duration, daily streak counts, or any metric that incentivises continued device use independent of skill transfer.
- Autonomously adapt stimulation amplitude, frequency, or duration in real time to maximise any neural or physiological metric.
- Infer, claim to detect, or reward the user for states corresponding to Tier 3 or Tier 4 of the contemplative measurability taxonomy: equanimity, compassion, insight into anatta, rigpa recognition, stream-entry, or any equivalent attainment.
- Play continuous audio feedback, soundscapes, or ambient sound changes contingent on positive state classification. The only audio event in Layer 1 is the distraction cue. Layer 2 ambient audio responds to gross agitation and dullness only, not to positive state achievement.
- Connect the Layer 3 stimulation current to any classifier output, threshold, or reward signal. Stimulation amplitude is set manually at session start and is not adjusted by software during a running session.

10.2 The Rationale for These Constraints

These constraints are not conservative caution. They are the direct engineering implication of the theoretical paper (Bose, 2026). Every constraint corresponds to a specific failure mode documented there: proxy reward hacking, inner alignment failure through performance mimicry or suppression, distributional shift through device dependency, or Tier 4 overclaiming. A device that violates any of these constraints is subject to the contemplative alignment problem in a documented, predictable way. This section is what makes the architecture alignment-aware rather than merely contemplative-themed.

12. Dependency Minimisation Protocol

Any successful contemplative cybernetic system should reduce long-term dependence on itself. Therefore all intervention systems within Sati-Loop are required to implement scaffold withdrawal schedules and transfer testing protocols. This is not an optional design feature. It is a defining property of the architecture and a necessary condition for claiming that the system develops genuine contemplative skill rather than device-dependent performance.

11.1 The Dependency Test

At any point during or after the training protocol, the system can be evaluated for dependency by running a device-absent session and measuring Delta t. If Delta t in the device-absent session is not significantly lower than the Phase A baseline, the training has not produced transferable skill. The device has produced a performance that requires the device to be reproduced. This is dependency, not skill, and it is a failure of the system regardless of how well the user performs in device-present sessions.

11.2 Withdrawal Schedule

- The classifier threshold T and the minimum cue interval are automatically adjusted upward as session-mean Delta t decreases, according to the schedule defined in Section 4.5.
- Weekly device-absent transfer sessions are mandatory from Week 3 onward. They cannot be disabled by the user.
- The app does not display a transfer test score or reward the user for good transfer performance. Transfer data is logged privately for research purposes and displayed only as a trend line showing whether unassisted Delta t is moving in the correct direction over time.
- If the transfer test shows no improvement after eight weeks of active training, the system does not extend the scaffolding. It flags the stall and recommends the user discuss their practice with a teacher. Technology cannot substitute for the diagnostic insight of an experienced practitioner.

11.3 Contribution Validation Rules

To preserve the intellectual identity of the project and prevent drift toward commercial health-hacking patterns, all pull requests and community hardware modifications must satisfy three validation criteria before integration into the main branch:

- **Subtractive feedback verification.** Any modification that introduces persistent visual metrics, continuous audio feedback, gamified scoring, or session ratings is rejected. The output engine must remain silent during aligned states.
- **Device-absent transfer verification.** Any new classification feature or intervention modality must be validated against Phase C transfer testing. If a feature improves in-session performance but does not transfer to unassisted practice, it is rejected.
- **Modulation decoupling rule.** Any modification that links classifier probabilities to automatic adjustment of stimulation parameters (tDCS, tACS, or taVNS amplitude) is rejected. Stimulation is restricted to Layer 3 manual-floor, emergency-override configurations as specified in Section 5.3.

14. Comparison with Existing Systems

System	Sensing	Feedback type	Transfer test	Alignment aware	Open source	Cost
Muse S	EEG (4ch)	Continuous audio (calm reward)	No	No	No	~USD 380
Unyte IOM2	HRV only	Gamified resonance score	No	No	No	~USD 140 + sub

System	Sensing	Feedback type	Transfer test	Alignment aware	Open source	Cost
HeartMath Inner Balance	HRV only	Coherence score	No	No	No	~USD 130
Clinical neurofeedback	EEG (full)	Visual reward for target frequency	Rarely	No	No	USD 2,000+
OpenBCI (raw)	EEG (8-16ch)	None (raw research tool)	N/A	N/A	Yes	~USD 1,000
Sati-Loop v1	EEG + HRV + resp + IMU	Minimal distraction cue only; no reward	Yes (primary metric)	Yes (by design)	Yes	~USD 450-1,300

15. Development Roadmap

Version	Target	Key additions
v1.0 (this spec)	Research platform	Core sensing, Layer 1 cueing, transfer test protocol, open hardware spec
v1.1	Software complete	BrainFlow pipeline, RF classifier, scaffold withdrawal, calibration app
v1.2	Layer 2 active	HRV-guided breath pacing, state-responsive ambient audio
v2.0	Validated	Published pilot study results, revised classifier based on real data, GSR addition
v2.1	taVNS integration	Layer 3 hardware spec, isolation PCB, safety protocol, pilot study
v3.0	Platform	Open dataset release, community protocols, multi-tradition support

16. Citation and License

12.1 How to Cite

If you build on this specification, please cite both the design specification and the theoretical paper:

Bose, J. (2026). Sati-Loop: Design Specification for an Alignment-Aware Contemplative Cybernetic Platform, v1.0. GitHub. <https://github.com/joybosero/sati-loop>

Bose, J. (2026). The Contemplative Alignment Problem: Reward Misspecification in Closed-Loop Meditation Systems. PsyArXiv. <https://osf.io/preprints/psyarxiv/>

12.2 License

- Hardware designs (schematics, PCB files, STL files): CERN Open Hardware Licence v2, Strongly Reciprocal (CERN-OHL-S). Any derivative hardware must be released under the same licence.
- Software and firmware: MIT Licence. Use freely with attribution.
- Documentation: Creative Commons Attribution 4.0 International (CC BY 4.0).

The use of the name Sati-Loop for any system that rewards proxy contemplative states, claims to detect Tier 3 or Tier 4 attainments, or includes neural stimulation connected to the classifier reward loop is explicitly prohibited under the terms of this licence.

Appendix A: Glossary

Term	Definition
Attentional scaffold	A technological system that supports the development of a skill but is designed to be withdrawn as the skill matures.
Contemplative alignment problem	The structural risk in closed-loop meditation devices that the practitioner's brain will optimise for the measurable proxy signal rather than the intended contemplative capacity. Formalised in Bose (2026).
Delta t	The interval between distraction onset (detectable by EEG) and conscious self-correction by the practitioner. The primary outcome measure for Sati-Loop.
Goodhart's Law	When a measure becomes a target, it ceases to be a good measure. The foundational principle behind the contemplative alignment problem.
Inner alignment	Whether the practitioner's learned strategy corresponds to the intended contemplative mechanism, rather than a shortcut that produces the target signal.
Outer alignment	Whether the measurable proxy (e.g. calm EEG) actually captures the intended contemplative target (e.g. genuine equanimity).
Proxy reward	A measurable signal used as a substitute for the true target in a feedback loop. The source of most alignment failures in contemplative neurotechnology.
Scaffold withdrawal	The systematic reduction of device intervention as the practitioner's skill develops. A design requirement in Sati-Loop, not an optional feature.
taVNS	Transcutaneous auricular vagus nerve stimulation. Non-invasive stimulation of the vagal branch at the ear tragus, which upregulates parasympathetic tone via the brainstem. Layer 3 of Sati-Loop.
Tier 1-4 states	The four-tier taxonomy of contemplative measurability from Bose (2026). Tier 1: reliably measurable now. Tier 2: candidate correlates. Tier 3: currently unmeasurable. Tier 4: possibly structurally unmeasurable in third-person terms.
Transfer test	A session conducted without the device, used to determine whether skill improvements persist in unassisted practice. The only success criterion in Sati-Loop.

End of Sati-Loop Design Specification v1.0

